Artifact Removal Factor for Circular-view Photoacoustic Tomography

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Abstract— **Reducing the number of acoustic detectors in a circular-view photoacoustic imaging system leads to cost reduction and increased efficiency. But reducing the number of acoustic detectors results in appearing more streak artifacts in the reconstructed image. In this paper to suppress artifacts, we proposed an artifact removal factor. The proposed factor is a combination of an edge detection image filter and a standard deviation filter that is applied to the back-projected signals. To evaluate the performance of the proposed method, we reduced the number of transducers to a fifth of the minimum required number. Numerical results have shown that the generalized contrast to noise ratio (gCNR) reaches 0.85 in the proposed method. Also, the degradation of the gCNR is not significant by reducing the number of acoustic detectors from 1200 to 240. Moreover, the structural similarity index measure (SSIM) values reach 0.91 and 0.94 for 1200 and 240 acoustic detectors, respectively, which is more than ~100% improvement compared to the conventional method.**

Keywords—Photoacoustic, Circular full-view, Streak artifact, Tomography..

I. Introduction

In the circular full-view configuration of a photoacoustic (PA) imaging (C-PAI) system, more solid angles of a generated PA wave are captured by detectors [1, 2]. This results in obtaining a higher quality reconstructed image in terms of resolution compared to planar and circular limited-view configurations [3]. In two circumstances, the quality of reconstructed image in a C-PAI system may be degraded; 1) if detectors are of a large aperture, the resolution degrades at off-center distances [1, 4]. 2) reducing the number of acoustic detection points results in appearing streak artifacts and aliasing [5, 6].

To compensate for the effect of a large aperture detector, small aperture detectors, and ring-shaped arrays are developed for a C-PAI system [7]. Some advanced reconstruction algorithms are presented to improve the resolution in the off-center distances [8, 9]. Also, positive and negative focused transducers are used to compensate for the effect of a large aperture detector [10]. A new imaging setup for the C-PAI system based on the virtual detector concept is developed to improve resolution in the off-center distances [11, 12].

Several studies have been proposed to remove artifacts. Hu, et al. [6] proposed a spatiotemporal antialiasing algorithm to mitigate streak artifacts in the PAI system. Kalva and Pramanik [13] proposed a modified delay-and-sum reconstruction algorithm to improve resolution and suppress artifacts. Some advanced reconstruction algorithms have been presented to suppress artifacts in a C-PAI system [14-16]. The signal restoration algorithm is developed to improve the signal-to-noise ratio (SNR) of the reconstructed image [14, 15]. Deep learning-based methods are also presented extensively to increase contrast ratio and image quality [17-19]. Although deep learning algorithms improve the quality of the reconstructed image, the lack of large experimental datasets limits their performance in *in vivo* scenarios. In other works, some iterative image and compress sensing-based methods are also proposed to improve the image quality [20, 21]. Awasthi, et al. [22] offered image-guided filtering for improving image reconstruction. Some weighting factors are also presented to remove artifacts and eliminate noise [e.g., the Coherence factor (CF)] [23]. However, CF causes missing data in the reconstructed image [24, 25].

In this paper, a new weighting factor is proposed to remove streak artifacts in a C-PAI system. The proposed method consists of a two stage factor. The first stage is the standard deviation factor that should be applied to the back-projected signals from detectors. The second stage is an edge detection factor which is applied to the initial reconstructed image. By digitalizing the output of multiplication of two factors, and multiplying the new factor to the reconstructed image, streak artifacts will be eliminated and a high-quality reconstructed image will be obtained.

II. Materials and Methods

Although the optical absorption coefficient of each PA absorber is positive, the generated PA signal would be bipolar [see Fig. 1(a)]. Since the detectors are not directional, the back-projection signal in the reconstruction process should be arc-shaped. The reconstructed value of each pixel in the region of interest (ROI) is a summation of the related back-projected value of all transducers. If a pixel in the ROI is related to the presence of an excited PA absorber, all back-projected signals will have non-zero (positive or negative) values at that location [see the red dot in Fig. 1(b)]. But, if a pixel refers to an artifact, some of the back-projected signals may have zero values at that location [see green and orange dots in Fig. 1(b)]. Due to the high number of zeros, the standard deviation value of the back-projected signals at that pixel would be low. Since the PA signal is symmetrical, at each pixel of the ROI, related to the presence of a PA absorber, the back-projected values of each two transducers that face each other are of opposite signs. Therefore, the standard deviation value of pixels related to the PA absorber [e.g., red dot in Fig. 1(b)] is higher than those related to the artifacts [e.g., green and orange dots in Fig. 1(b)]. Therefore, this standard deviation factor extracts the locations that refer to the excited PA absorbers and artifacts.

2022 IEEE International Ultrasonics Symposium (IUS) | 978-1-6654-6657-8/22/$31.00 ©2022 IEEE | DOI: 10.1109/IUS54386.2022.9958228

This standard deviation filter is the first stage of our proposed factor. In the standard deviation factor, the location of some artifacts may be mistakenly detected as the absorber location [e.g., A2 in Fig. 1(b)]. This issue may happen more in crowded ROIs. Therefore, a second stage filter should be needed to separate more areas. For more accuracy in separating the locations of PA absorbers and artifacts, a digitalized edge detector filter is applied to the reconstructed image as the second stage of the proposed filter. The proposed factor is a product of two-stage filter. Applying both stages of the proposed factor to the reconstructed image will suppress streak artifacts. This artifact suppression is proven by numerical studies in the following section.

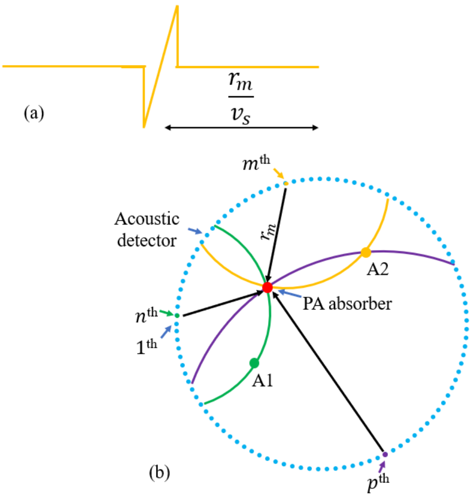

Fig. 1. (a) schematic of a captured PA signal by an infinite bandwidth acoustic detector. r_m is distance between excited PA absorber and v_s is the speed of sound in the medium. (b) schematic of a C-PAI system with n acoustic detectors. A1 and A2 represent the location of artifacts.

III. Numerical Study and Results

A. Numerical Study

k-Wave MATLAB toolbox was used to synthesize the numerical raw data [26]. The imaging region and grid sizes were 2000 and 0.025 mm in both X-Y directions. The radius of the imaging region was 5 cm. Also, the speed of sound was considered 1500 m/s. The central frequency and bandwidth of acoustic detectors were 2.25 MHz and 70%. In a C-PAI system, the required minimum acoustic detectors (N_{DP}) are given as follows [6]:

$$N_{DP} = \frac{8\pi R}{\lambda} \tag{1}$$

where, R is radius of the imaging region, and λ is central wavelength of the acoustic detector. Therefore, the required minimum number of acoustic detectors should be 1200. To evaluate the performance of the proposed method in suppressing artifacts, three numerical studies with a different number of acoustic detectors (1200, 480, and 240) were conducted. To compare the results of the presented method with existing methods, the data synthesized by three methods, universal back-projection (UBP) [27], interpolation-based [6], and contamination-tracing back-projection (CTBP) [5], have also been reconstructed. The schematic of the numerical phantom and imaging region are shown in amplitude and dB scale in Figs. 2(a, b), respectively. Also, for a quantitative comparison between different methods (different reconstruction algorithms and different number of acoustic detectors), two quality indices including structural similarity index measure (SSIM) and generalized contrast-to-noise ratio (gCNR) [28] were measured for all 12 different cases.

B. Numerical Results

The numerical results of different reconstruction methods are presented in Fig. 2 in a dynamic range of 60 dB. In the UBP method, as the number of detectors decreases, more streak artifacts appear. When the number of detectors is reduced to 20% of the required value (240), the quality of the image drops drastically, and aliasing and streak artifacts strongly appear [see Fig. 2(b1)]. Quantitatively speaking, the SSIM value decreases from 0.80 to 0.42 by reducing the number of detectors from 1200 to 240. The interpolation method suppresses more streak artifacts in 240 detectors case compared to the UBP method. The interpolation method improves SSIM by ~30%. The results of the interpolation method are shown in Fig. 2(c). The CTBP method can further suppress more artifacts in the reconstructed image, especially in the 240 detection points case. The CTBP method improves the SSIM values for 240 detection points case by ~100%. To achieve the best results, some variables should be tuned in the CTBP method. Therefore, α, w_{min} and the threshold values were fixed at 0.3, 0.15, and 0.5, respectively. Although the SSIM values are improved in interpolation and CTBP methods, the gCNR values are degraded. The results of the CTBP method are presented in Fig. 2(d). The proposed method improves both the SSIM and gCNR values. The proposed method improves the SSIM values by up to 115%. The SSIM value for 240 detectors reaches 0.91 in the proposed method, which is more than the SSIM value for 1200 detectors in the UBP method. Moreover, the proposed method improves gCNR values by up to ~15% compared to UBP. The quantitative values of the SSIM and gCNR for different methods are revealed in TABLE I and II.

TABLE I. Quantitative Values of SSIM

Method	Number of detection points		
	240	***480***	***1200***
UBP	0.42	0.80	0.85
CTBP	0.83	0.90	0.90
Interpolation	0.57	0.84	0.86
Proposed method	**0.91**	**0.94**	**0.94**

TABLE II. Quantitative Values of gCNR

Method	Number of detection points		
	240	***480***	***1200***
UBP	0.72	0.80	0.80
CTBP	0.55	0.58	0.59
Interpolation	0.65	0.76	0.79
Proposed method	**0.83**	**0.86**	**0.85**

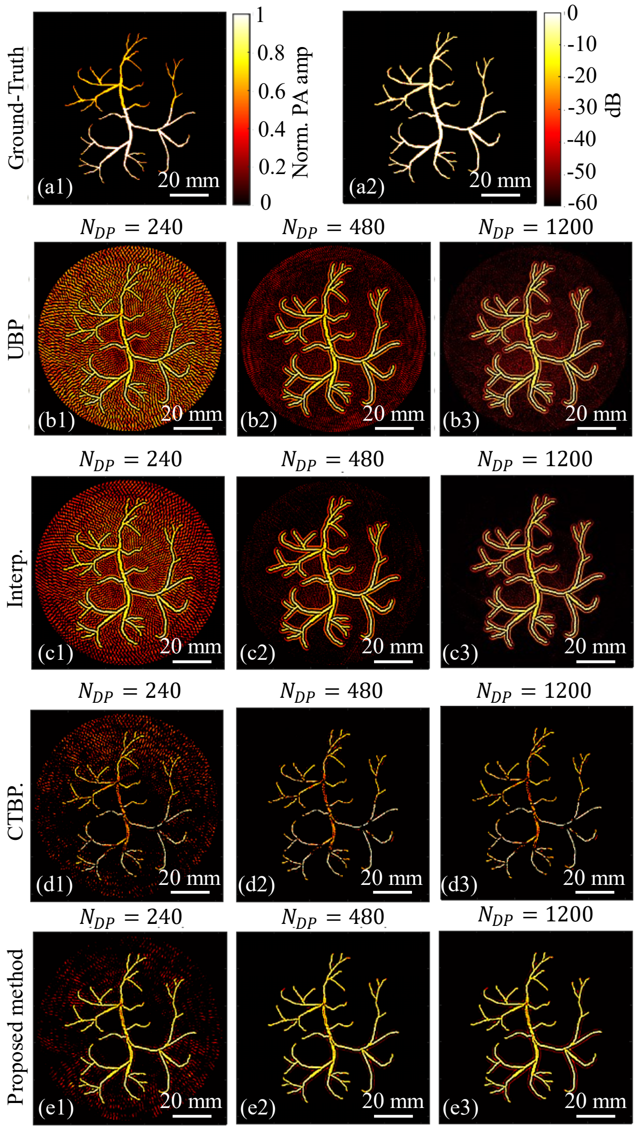

Fig. 2. (a1, a2) synthesized phantom in amplitude and dB scales. (b-e) reconstructed image by UBP, interpolation, CTBP, and the proposed methods, respectively. The N_{DP} value indicates the number of acoustic detectors used in each study.

IV. Conclusion

In this paper, to suppress artifacts in a C-PAI system, an artifact removal factor has been presented. The proposed method is a combination of two stage factors. One factor is applied to back-projected signals and the other one is applied to the initial reconstructed image. The proposed method has been evaluated via numerical studies. For the number of acoustic detectors 1200/240, the gNCR values were measured at 0.85/0.83 and 0.8/0.72 for the proposed and conventional methods, respectively. It can be concluded that the degradation of the gCNR values of the proposed method is not significant by reducing the number of acoustic detectors from 1200 to 240. Also, the SSIM values for the number of acoustic detectors of 1200/240, were measured at 0.94/0.91 and 0.85/0.42 for the proposed and conventional methods, respectively. Moreover, the proposed method can be combined with coprime sparse array- based methods for further improvement in the future study [29].

References

[1] M. Xu and L. V. Wang, "Analytic explanation of spatial resolution related to bandwidth and detector aperture size in thermoacoustic or photoacoustic reconstruction," *Physical Review E,* vol. 67, no. 5, 2003, doi: 10.1103/physreve.67.056605.

[2] D. Das, A. Sharma, P. Rajendran, and M. Pramanik, "Another decade of photoacoustic imaging," *Phys Med Biol,* Dec 23 2020, doi: 10.1088/1361-6560/abd669.

[3] S. Hakakzadeh and Z. Kavehvash, "Blind Angle and Angular Range Detection in Planar and Limited-View Geometries for Photoacoustic Tomography," in *2021 29th Iranian Conference on Electrical Engineering (ICEE)*, 18-20 May 2021 2021, pp. 922-926, doi: 10.1109/ICEE52715.2021.9544114.

[4] S. Hakakzadeh and Z. Kavehvash, "Image Quality Equations for Focused Transducer in Circular Photoacoustic Computed Tomography," in *2021 29th Iranian Conference on Electrical Engineering (ICEE)*, 18-20 May 2021 2021, pp. 917-921, doi: 10.1109/ICEE52715.2021.9544449.

[5] C. Cai, X. Wang, K. Si, J. Qian, J. Luo, and C. Ma, "Streak artifact suppression in photoacoustic computed tomography using adaptive back projection," *Biomed Opt Express,* vol. 10, no. 9, pp. 4803-4814, Sep 1 2019, doi: 10.1364/BOE.10.004803.

[6] P. Hu, L. Li, L. Lin, and L. V. Wang, "Spatiotemporal Antialiasing in Photoacoustic Computed Tomography," *IEEE Trans Med Imaging,* vol. 39, no. 11, pp. 3535-3547, Nov 2020, doi: 10.1109/TMI.2020.2998509.

[7] L. Li *et al.*, "Single-impulse panoramic photoacoustic computed tomography of small-animal whole-body dynamics at high spatiotemporal resolution," *Nature Biomedical Engineering,* vol. 1, no. 5, p. 0071, 2017/05/10 2017, doi: 10.1038/s41551-017-0071.

[8] B. Wang, T. Ye, G. Wang, L. Guo, and J. Xiao, "Approximate back-projection method for improving lateral resolution in circular-scanning-based photoacoustic tomography," *Med Phys,* Apr 10 2021, doi: 10.1002/mp.14880.

[9] J. Xiao, X. Luo, K. Peng, and B. Wang, "Improved back-projection method for circular-scanning-based photoacoustic tomography with improved tangential resolution," *Appl Opt,* vol. 56, no. 32, pp. 8983-8990, Nov 10 2017, doi: 10.1364/AO.56.008983.

[10] L. Nie, Z. Guo, and L. V. Wang, "Photoacoustic tomography of monkey brain using virtual point ultrasonic transducers," *J Biomed Opt,* vol. 16, no. 7, p. 076005, Jul 2011, doi: 10.1117/1.3595842.

[11] S. Hakakzadeh *et al.*, "Multi-angle data acquisition to compensate transducer finite size in photoacoustic tomography," *Photoacoustics,* vol. 27, p. 100373, 2022/09/01/ 2022, doi: https://doi.org/10.1016/j.pacs.2022.100373.

[12] S. Hakakzadeh *et al.*, "Finite Transducer Size Compensation in Two-Dimensional Photoacoustic Computed Tomography," in *2021 IEEE International Ultrasonics Symposium (IUS)*, 11-16 Sept. 2021 2021, pp. 1-3, doi: 10.1109/IUS52206.2021.9593757.

[13] S. K. Kalva and M. Pramanik, "Experimental validation of tangential resolution improvement in photoacoustic tomography using modified delay-and-sum reconstruction algorithm," *Journal of Biomedical Optics,* vol. 21, no. 8, p. 086011, 2016. [Online]. Available: https://doi.org/10.1117/1.JBO.21.8.086011.

[14] Soheil Hakakzadeh, Y. Z. Mohammadreza Amjadian, Seyed Masood Mostafavi, Zahra Kavehvash, and Lidai Wang, "Signal Restoration Algorithm for Photoacoustic Imaging Sytems" *IEEE TUFFC,* 2022.

[15] Soheil Hakakzadeh, Seyed Masood Mostafavi, and Zahra Kavehvash, "Unipolar Back-Projection Algorithm for Photoacoustic Tomography," *IUS,* 2022.

[16] S. Hakakzadeh, S. M. Mostafavi, and Z. Kavehvash, "Fast Adapted Delay and Sum Reconstruction Algorithm in Circular Photoacoustic Tomography," in *2022 30th International Conference on Electrical Engineering (ICEE)*, 17-19 May 2022 2022, pp. 49-54, doi: 10.1109/ICEE55646.2022.9827413.

[17] N. Davoudi, X. L. Deán-Ben, and D. Razansky, "Deep learning optoacoustic tomography with sparse data," *Nature Machine Intelligence,* vol. 1, no. 10, pp. 453-460, 2019, doi: 10.1038/s42256-019-0095-3.

[18] P. Rajendran and M. Pramanik, "Deep learning approach to improve tangential resolution in photoacoustic tomography," *Biomed Opt Express,* vol. 11, no. 12, pp. 7311-7323, Dec 1 2020, doi: 10.1364/BOE.410145.

[19] S. Gutta, V. S. Kadimesetty, S. K. Kalva, M. Pramanik, S. Ganapathy, and P. K. Yalavarthy, "Deep neural network-based bandwidth enhancement of photoacoustic data," *J Biomed Opt,* vol. 22, no. 11, pp. 1-7, Nov 2017, doi: 10.1117/1.JBO.22.11.116001.

[20] N. Awasthi, S. Kumar Kalva, M. Pramanik, and P. K. Yalavarthy, "Dimensionality reduced plug and play priors for improving photoacoustic tomographic imaging with limited noisy data," *Biomed Opt Express,* vol. 12, no. 3, pp. 1320-1338, Mar 1 2021, doi: 10.1364/BOE.415182.

[21] A. Ghanbarzadeh-Dagheyan *et al.*, "Time-domain ultrasound as prior information for frequency-domain compressive ultrasound for intravascular cell detection: A 2-cell numerical model," *Ultrasonics,* vol. 125, p. 106791, Sep 2022, doi: 10.1016/j.ultras.2022.106791.

[22] N. Awasthi, S. K. Kalva, M. Pramanik, and P. K. Yalavarthy, "Image-guided filtering for improving photoacoustic tomographic image reconstruction," *Journal of biomedical optics,* vol. 23, no. 9, p. 091413, 2018.

[23] S. Hakakzadeh, S. M. Mostafavi, M. Amjadian, and Z. Kavehvash, "Adapted Coherent Weighting in Photoacoustic Tomography," in *2021 28th National and 6th International Iranian Conference on Biomedical Engineering (ICBME)*, 25-26 Nov. 2021 2021, pp. 28-32, doi: 10.1109/ICBME54433.2021.9750310.

[24] Soheil Hakakzadeh and Zahra Kavehvash, "A Hilbert-based Coherence Factor for Photoacoustic Imaging," *IEEE TUFFC,* 2022.

[25] O. M. H. Rindal, A. Austeng, A. Fatemi, and A. Rodriguez-Molares, "The Effect of Dynamic Range Alterations in the Estimation of Contrast," *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control,* vol. 66, no. 7, pp. 1198-1208, 2019, doi: 10.1109/tuffc.2019.2911267.

[26] B. Treeby and B. Cox, "k-Wave: MATLAB toolbox for the simulation and reconstruction of photoacoustic wave fields," *Journal of Biomedical Optics,* vol. 15, no. 2, p. 021314, 2010. [Online]. Available: https://doi.org/10.1117/1.3360308.

[27] M. Xu and L. V. Wang, "Universal back-projection algorithm for photoacoustic computed tomography," *Physical Review E,* vol. 71, no. 1, 2005, doi: 10.1103/physreve.71.016706.

[28] K. M. Kempski, M. T. Graham, M. R. Gubbi, T. Palmer, and M. A. Lediju Bell, "Application of the generalized contrast-to-noise ratio to assess photoacoustic image quality," *Biomed Opt Express,* vol. 11, no. 7, pp. 3684-3698, Jul 1 2020, doi: 10.1364/BOE.391026.

[29] Vahid Amin Nili, Mina Ezati, Yan Yan, Zahra Kavehvash, and Mohammad Mehrmohammadi, "Field of View and Resolution Improvement in Coprime Sparse Synthetic Aperture Ultrasound Imaging," 2022: IEEE.